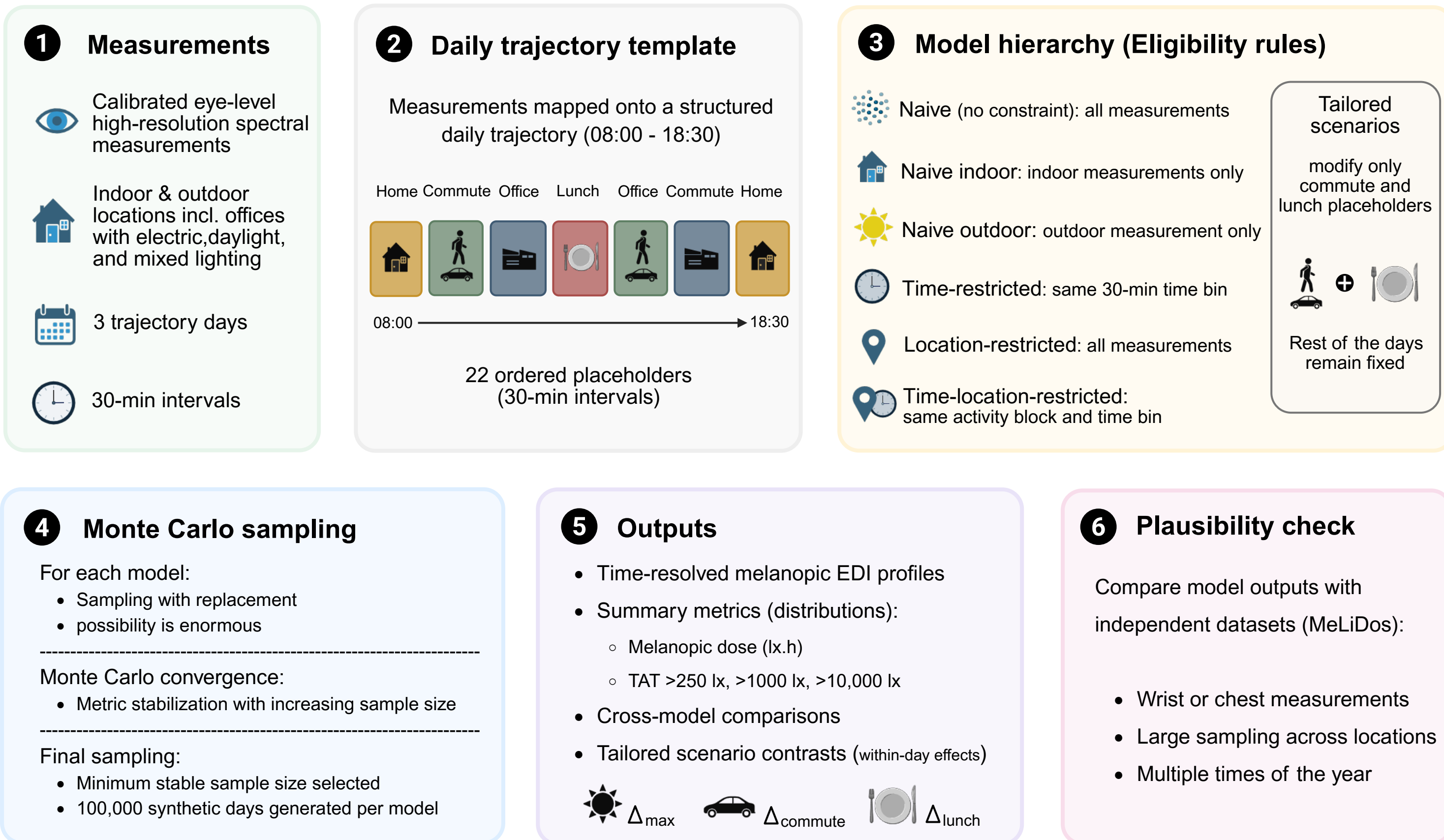
1 Measurements
Calibrated eye-level high-resolution spectral measurements
Indoor & outdoor locations incl. offices with electric,daylight, and mixed lighting
3 trajectory days
30-min intervals
2 Daily trajectory template
Measurements mapped onto a structured daily trajectory (08:00 - 18:30)
Home Commute Office Lunch Office Commute Home
08:00
18:30
22 ordered placeholders (30-min intervals)
3 Model hierarchy (Eligibility rules)
Naive (no constraint): all measurements
Naive indoor: indoor measurements only
Naive outdoor: outdoor measurement only
Time-restricted: same 30-min time bin
Location-restricted: all measurements
Time-location-restricted: same activity block and time bin
Tailored scenarios
modify only commute and lunch placeholders
Rest of the days remain fixed
4 Monte Carlo sampling
For each model:
Sampling with replacement
possibility is enormous
Monte Carlo convergence:
Metric stabilization with increasing sample size
Final sampling:
Minimum stable sample size selected
100,000 synthetic days generated per model
5 Outputs
Time-resolved melanopic EDI profiles
Summary metrics (distributions):
Melanopic dose (lx.h)
TAT >250 lx, >1000 lx, >10,000 lx
Cross-model comparisons
Tailored scenario contrasts (within-day effects)
Δ_{max}
$\Delta_{commute}$
Δ_{lunch}
6 Plausibility check
Compare model outputs with independent datasets (MeLiDos):
Wrist or chest measurements
Large sampling across locations
Multiple times of the year